ArduPilot Implementation of Optimal Control for Tilt Rotor UAV Landing

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Bachelor of Engineering (Honours)

by
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This thesis contains no material which has been accepted for the award of any other degree or diploma in any university. To the best of the author's knowledge, it contains no material previously published or written by another person, except where due reference is made in the text.

Lincoln Karavas
2 November 2026

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Abstract

This project puts forward an implementation of optimal control in ArduPilot to reduce the energy consumption of tilt rotor Unmanned Aerial Vehicles (UAVs) during landing by 61% compared to the current ArduPilot approach, with no hardware changes required. This is achieved through the use of a time-varying linear quadratic regulator control approach and a precomputed optimised trajectory. The goal of this project is to eventually expand this work to all powered-lift aircraft currently running ArduPilot. The other contribution of this work is the introduction of momentum drag into the ArduPilot simulation environment software in the loop (SITL), allowing users to simulate new manoeuvres utilising these effects or have more accurate simulation and flight planning.

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Nomenclature

SOS	Save Our Souls

Introduction

This project puts forth a method and implementation for ardupilot of optimal control for tilt rotor Unmanned Aerial Vehicles (UAV) landing transitions (specifically the Arace Griffin Pro) to reduce the energy consumption and landing period by 61% and 51% respectivly compared to the stock ArduPilot approach. As the use of small and medium scale powered lift UAVs increases in commercial settings so too does the demand for greater endurance and efficiency. This is generally approached from a hardware perspective, through higher density batteries, more efficient propulsion or lighter and more efficient air frames. However this project attempts to solve this through exploiting the dynamics of the aircrafts to reduce control effort, through a optimal control approach. Trajectory and control optimisations for UAVs is not a new area of research, however, this project provides an new and easier way to implement this for end users, as the control is integrated into ArduPilot, an existing and widespread approach to autonomous control of UAVs. This project specifically looks at the Arace Griffin Pro tilt rotor. This thesis starts by putting forth a longitudinal dynamical model of tilt rotor aircraft, by combining previous models of multirotor and fixed-wing aircraft. Followed by a method of trajectory optimisation, control and simulation.

Powered lift UAVs are a class of aircraft that have two separate flight modes. The first being fixed-wing flight like that of a conventional plane, the second being hover, similar to a multi-rotor. The combination of these two modes mean that the aircraft can takeoff and land without specialised infrastructure such as a runway, while being able to cruise in the more efficient fixed-wing flight mode, as the rotors in the fixed-wing mode only have to impart enough energy to balance the drag force, while in hover the rotors have to overcome the gravitational force of the entire airframe. Powered lift UAVs achieve this in different ways, some have separate propellers for hover and fixed-wing flight, some tilt their propellers from vertical to horizontal and some even tilt the entire wing with propellers statically mounted. These different airframe types all different approaches to transitioning between flight modes, but this project specifically looks at the Arace Griffin Pro a tilt rotor aircraft (Figure 1.1).



Figure 1.1: Arace Griffin Pro.

Literature Review

This chapter reviews the literature relevant to the energy optimised back-transition of a tilt rotor UAV. It begins with powered lift aircraft and the transition problem, then covers the control approaches that have been applied to it, followed by the trajectory optimisation methods used to generate transition manoeuvres. Post-stall manoeuvres is then reviewed as it is the closest analogue to the trajectory produced in this project, followed by trajectory tracking with time-varying LQR, the modelling of rotor forces in non-axial flow, and finally the implementation of controllers on open-source autopilots. The chapter concludes by identifying the gap this thesis addresses.

2.1 Powered Lift Aircraft and the Back-Transition

Powered Lift UAVs combine the vertical take off and landing capability of a multi rotor with the efficient forward flight of a fixed wing aircraft [12, 26]. The downside of this is the complicated flight mode achieved during the transition between the two. Ducard [12], provide a review of the different air frame types and applicable control architectures. They categorised all powered lift aircraft into, tail sitters; an aircraft that pitches to vertical to perform vertical take offs and landings, tilt rotors; where either all or a subset of the rotors tilt from vertical to forward, tilt wings; similar to tilt rotors however, the wing tilts with the rotor and quadplanes; that have separate propulsion systems for hover and fixed wing flight. The aircraft used in this project is the Griffin Pro which is a tilt rotor, where only the front two rotors of the quad tilt, and the model used is a modified version of that presented by Miller [22].

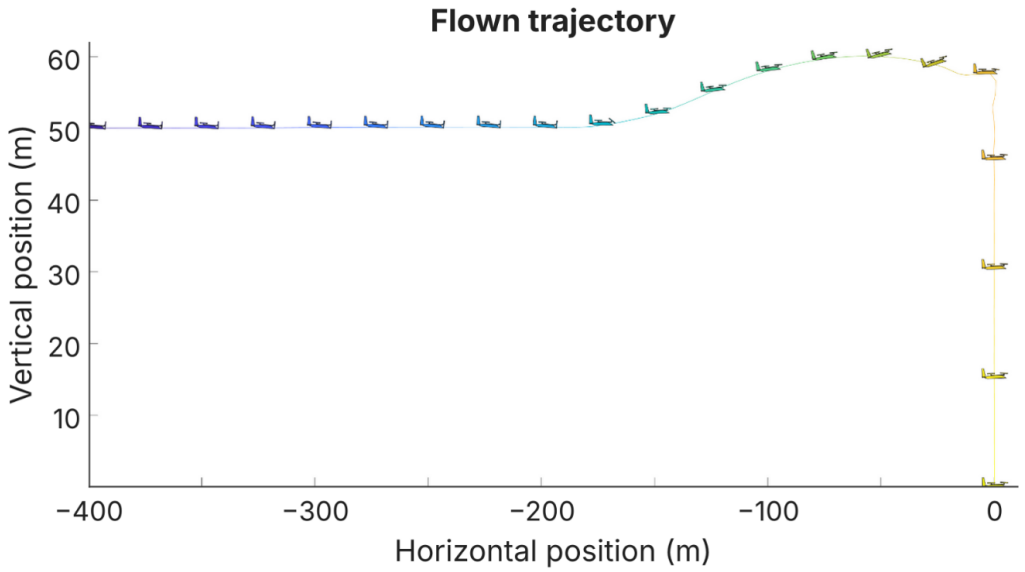


Figure 2.1: ArduPilot Landing Approach

The transition from hover to fixed wing flight has been extensively studied as there is a much greater risk for instability of the systems as the aircraft is in a lower energy state leading to unrecoverable stalls. The transition from fixed wing flight to hover is less studied as there is a obvious and stable solution of reconfiguring the aircraft for hover while coasting in fixed wing cruise. This is the approach used in current solutions such as ArduPilot. However this approach results in extensive time spent in hover, a substantially more power intensive flight mode compared to fixed wing flight (ref Figure (4.2)), as the active rotors have to counteract the gravitational force rather than the passive wings. The obvious solution of descending quicker in hover to minimise time spent in that mode has its own issues, primarily entering the vortex ring state. A phenomenon where if a rotor powered aircraft descends at a comparable rate to the downwash velocity produced by the rotor, large vortices can form centered around the mid point of the rotor blade span and cause a dramatic loss in lift [9].

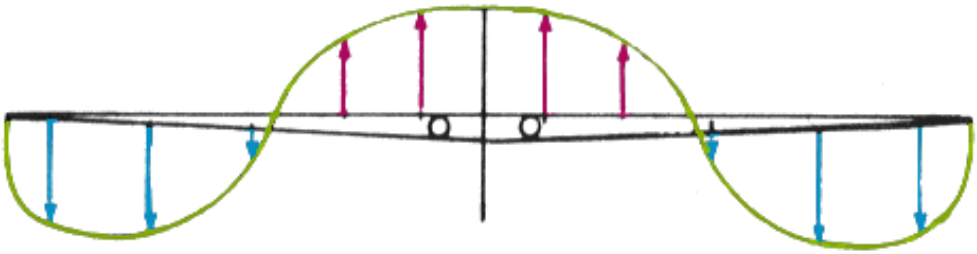


FIGURE 2-80. INDUCED FLOW VELOCITY DURING VORTEX RING STATE.

Figure 2.2: Vortex Ring State Air Flow

2.2 Transition Control

The simplest approach to transition control, and the one used by most open-source autopilots, is a scheduled transition, where the tilt angle and control mixing are blended as functions of airspeed or time and the attitude is held by PID loops [4]. This approach is robust and simple to tune, but it is not optimal with respect to any performance measure, and the trajectory the aircraft follows is a consequence of the schedule rather than a design choice.

Model based controllers have been proposed to improve on this. Allenspach and Ducard [1] applied nonlinear model predictive control (NMPC) to a tilt rotor UAV, controlling the aircraft for the entirety of flight utilising a single controller, and validated it in flight. Ducard and Allenspach [12] propose that this single controller approach should be the direction the field moves in, along with the need to manage control complexity against the computing resources available on board. The latter is a significant constraint for NMPC, which requires an optimisation problem to be solved in real time on board the aircraft's flight controller, and is a primary motivation for the precomputed controller used in this project.

Learning based approaches have also been applied. Fu et al. [14] used safe reinforcement learning for the back-transition of a ducted fan tail-sitter, minimising altitude change and transition time subject to a vertical velocity constraint, and showed that the learned trajectory closely matched the

optimal trajectory obtained with GPOPS-II.

2.3 Optimal Control

Trajectory optimisation refers to methods that find the control inputs which take a dynamic system between two states while minimising a cost function and satisfying constraints. Direct methods transcribe this infinite dimensional problem into a finite dimensional nonlinear program (NLP) by discretising the states and controls in time. Hargraves and Paris [15] introduced direct collocation, in which the system dynamics are enforced as equality constraints between discrete points, and Betts [7] surveys the numerical methods used to solve such problems.

Kelly [16] provides a practical tutorial on direct collocation, covering trapezoidal and Hermite–Simpson collocation, initialisation, and mesh refinement, and released the MATLAB library OptimTraj [17] that implements these methods; OptimTraj is the library used in this project. Kelly notes that the NLPs produced by transcription are generally non-convex, so the solver returns a locally optimal solution which depends on the initial guess. The straight line initialisation and successive mesh refinement described in Section **??** are intended to manage this.

Trajectory optimisation has been applied to the transition problem by several authors. Naldi and Marconi [24] formulated minimum-time and minimum-energy transition manoeuvres for a class of V/STOL aircraft as mixed-integer nonlinear programs. Zhao et al. [28] computed a transition corridor for a tilt rotor VTOL UAV and solved the transition strategy as a constrained optimal control problem using the Gauss pseudospectral method, with a cost function built on transition time, attitude stability and control continuity. Deng et al. [11] extended the conventional trim based transition corridor to a three dimensional dynamic corridor in nacelle angle, velocity and angle of attack, and used pigeon-inspired optimisation to find a smooth transition trajectory within it. Like this project, both Deng et al. and Fu et al. use a longitudinal three degree of freedom model, justified by the symmetry of the aircraft during the transition.

The common thread through these works is that the transition is constrained to remain within a corridor of near trimmed flight, and the cost function prioritises smoothness, time and attitude. Energy is either absent from the cost or, in the case of Naldi and Marconi, considered for a generic V/STOL model rather than a tilt rotor back-transition. Allowing the optimisation to leave the trimmed corridor and enter stalled flight is what produces the energy saving in this project, which leads to the post-stall literature reviewed next.

2.4 Post-Stall Manoeuvring and Perching

The optimal trajectory found in this project dives to convert potential energy into kinetic energy, then pulls up into a stall, using the drag at high angle of attack to dissipate the majority of the aircraft's kinetic energy. This is closely related to the perching manoeuvre performed by birds, which has been studied for fixed wing UAVs.

Cory and Tedrake [10] demonstrated a fixed wing glider performing a perching manoeuvre using a model identified from motion capture data and a trajectory optimised for the post-stall regime, stabilised with a time-varying LQR. Moore, Cory and Tedrake [23] extended this work using LQR-Trees, a feedback motion planning method built on time-varying LQR around nominal trajectories, and achieved a 95% perching success rate over 147 flights on real hardware across a wide range of initial speeds. Their results show that post-stall manoeuvres can be executed reliably with a simple

airframe and a model based controller, without complex wing morphology or sensing, at least in the absence of significant wind gusts.

These works establish that deliberately stalling an aircraft is a viable way to dissipate energy, and that time-varying LQR about an optimised trajectory is a suitable way to stabilise it. The tilt rotor in this project has an advantage over the glider in that the rotors provide control authority that is independent of airspeed, which is used to stabilise the stall in the final stage of the approach.

2.5 Trajectory Tracking with Time-Varying LQR

The linear quadratic regulator (LQR) minimises a quadratic cost on state error and control effort, with the optimal gain given by the solution of a Riccati equation [2]. For a time-varying system linearised about a trajectory over a finite horizon, the gain is found by integrating the differential Riccati equation backwards in time from a terminal condition, which produces a time-varying gain matrix (TVLQR) [2, 27]. As the horizon is finite, the gains reduce towards the end of the trajectory as the remaining cost of a state error becomes small relative to the control cost. This explains the reduction in gain seen near the end of the approach in Figure **??**. The weighting matrices are commonly chosen using Bryson's rule [8], where each diagonal weight is the inverse square of the maximum acceptable value of the corresponding state or input.

TVLQR should be distinguished from classical gain scheduling, where independent controllers are designed at a set of frozen operating points and interpolated as a function of a scheduling variable such as airspeed [25]. Rugh and Shamma [25] note that stability at each frozen operating point does not in general guarantee stability of the scheduled system, unless the scheduling variable changes slowly. The same caution applies to the frozen time eigenvalues of a time-varying closed loop system, so the stability of the controller in this project is also assessed along the trajectory as a whole rather than from the frozen time eigenvalues alone.

TVLQR is well suited to embedded implementation because all of the computation is done offline, and the onboard controller only needs to look up a gain and compute a matrix vector product each cycle. This is in contrast to NMPC, and is the reason it was selected for this project.

2.6 Rotor Aerodynamics in Non-Axial Flow

When a rotor operates with a component of the freestream perpendicular to its axis, the advancing blade sees a higher airspeed than the retreating blade, producing an in-plane force opposing the motion (the H-force) and blade flapping, as described by momentum and blade element theory [19]. Martin and Salaün [21] showed that this rotor drag is significant even on small multirotors at low speed, with a force approximately proportional to the product of rotor speed and airspeed, and that it explains the accelerometer readings of a quadrotor in flight. Mahony, Kumar and Corke [20] include blade flapping and induced drag in their tutorial model of multirotor dynamics, and Bangura and Mahony [6] used momentum theory to model rotor thrust for multirotor control. Faessler et al. [13] showed that modelling rotor drag explicitly, rather than treating it as an unknown disturbance, significantly reduces the trajectory tracking error of a quadrotor.

The optimal trajectory in this project uses the rotors in this way, with the aircraft's velocity perpendicular to the rotor axes during the stall, so that rotor drag contributes to dissipating the kinetic energy. The public release of the ArduPilot simulator does not model this force, which motivated the rotor model developed in Section **??**.

2.7 Implementation on Open-Source Autopilots

ArduPilot is an open-source autopilot that supports tilt rotor and other hybrid VTOL aircraft through its QuadPlane firmware [4]. It provides a Lua scripting environment that allows custom code to read the vehicle state and override the outputs to the servos and motors without modifying the firmware [3], and a software in the loop (SITL) simulator that runs the unmodified flight code against a simulated aircraft [5]. Together these allow an advanced controller to be deployed on the existing flight controller of a production aircraft and tested in simulation first, which is the approach taken in this project. The limited memory and processing available to the Lua environment are the main constraints on the controller, and motivate both the choice of TVLQR and the reduction in the number of stored gain matrices described in Section **??**.

The accuracy of any model based controller depends on the aerodynamic model. The aerodynamic coefficients of the model in this project were re-estimated from flight logs of the Griffin Pro, following the established approach of aircraft system identification from flight data [18].

2.8 Summary and Research Gap

The literature shows that the transition of hybrid VTOL aircraft has been approached through scheduled PID control, model predictive control, reinforcement learning, and trajectory optimisation within a transition corridor. The majority of this work focuses on the forward transition, constrains the aircraft to near trimmed flight, and optimises for smoothness, time and attitude rather than energy. Separately, the perching literature shows that post-stall manoeuvres can be stabilised reliably with time-varying LQR, and the multirotor literature shows that rotor drag in non-axial flow is a significant force that can be modelled and exploited.

No work was found that combines these to find an energy optimal back-transition for a tilt rotor that uses both the post-stall drag of the wing and the in-plane drag of the rotors, and implements it on an unmodified production flight controller. This thesis addresses that gap by optimising the back-transition of the Griffin Pro for minimum rotor energy, tracking it with a TVLQR implemented in ArduPilot's Lua environment, and evaluating it in a SITL simulator modified to model rotor drag.

Methodology

3.1 Aircraft Dynamical Model

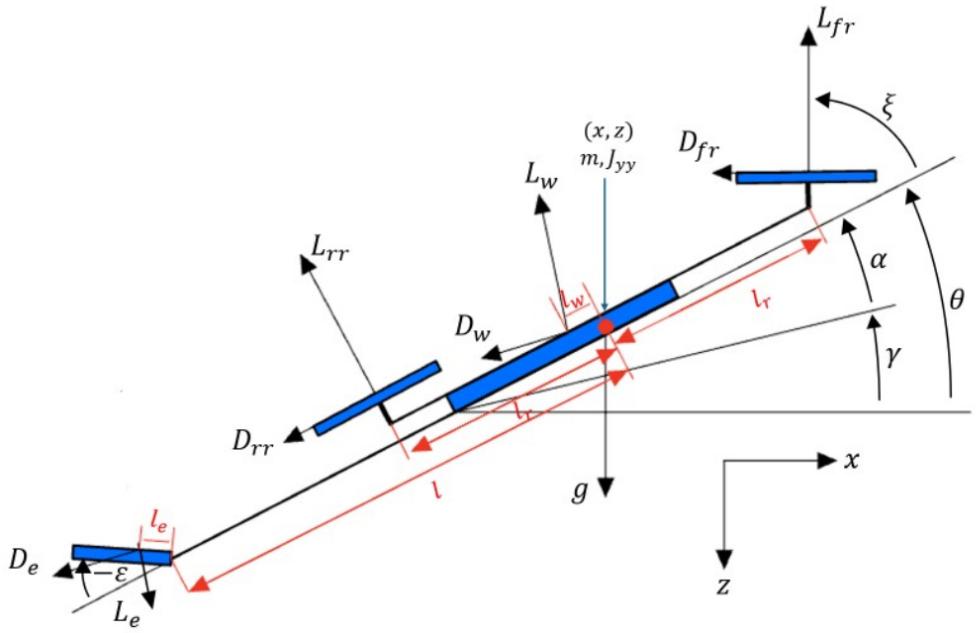


Figure 3.1: Dynamical Model Diagram

The Model used in this project is based of that presented in Miller's thesis [22]. As such the modeonly captures the longitundinal dynamics of the aircraft, this simplification was made as all of the behaviours being optimised are only present in the longitundinal case, while the lateral is being stabilised to trim level flight. The simplification to a two dimentional model means that the state space X can be simlified from $\mathbb{R}^{12}$ to $\mathbb{R}^6$, the input space(U) can also be simplified resulting in the spaces shown in (3.1)

$$X = \begin{bmatrix} x & z & \theta & \dot{x} & \dot{z} & \dot{\theta} \end{bmatrix}^T, \qquad U = \begin{bmatrix} \omega_f & \omega_r & \xi & \epsilon \end{bmatrix}^T \tag{3.1}$$

where

State		Input	
x	e_1 position (m)	ω_f	front rotor speed (rpm)
z	e_2 position (m)	ω_r	rear rotor speed (rpm)
θ	pitch (rad)	ξ	front rotor tilt (rad)
$\dot{x}$	e_1 velocity (m/s)	ϵ	elevator deflection (rad)
$\dot{z}$	e_2 velocity (m/s)		
$\dot{\theta}$	pitch rate (rad/s)		

Table 3.1: Model States and Inputs

The dynamics of the system are shown in (3.2) & (3.3), which are an application of Newton's second law of motion rearranged.

$$\begin{bmatrix} \ddot{x} \\ \ddot{z} \end{bmatrix} = \frac{1}{m}(R(\theta-\alpha)(F_w+F_e)) + R(\theta+\xi)F_{fr} + R(\theta)F_{rr} + mg) \tag{3.2}$$

$$\ddot{\theta} = \frac{1}{J_{yy}}(l_w R(-\alpha) + F_{w_\perp} + (lcos(\epsilon)+l_e)R(-\alpha)F_{e_\perp} - l_r R(\theta+\xi)F_{fr_\perp} + l_r R(\theta)F_{rr_\perp}) \tag{3.3}$$

where $R(\theta)$ is the standard two dimentaional rotation matrix.

3.1.1 Aerodynamic Surface Forces

The aerodynamic surface (wing & elevator) forces are modelled using a blended coefficents model to describe post-stall behaviour, as the standard approach of a linear model would mislead the trajectory optimisation when the aircraft would be in stall. The blended model approach utilises the standard linear airfoil coeffcient at angles of attack less than stall and smoothly transitions to a flat-plate model post-stall via a sigmoid function.

$$L_x = \frac{1}{2}s_x\rho|v_x^2|C_{l_x}(\alpha_x), \quad D_x = \frac{1}{2}s_x\rho|v_x^2|C_{d_x}(\alpha_x) \tag{3.4}$$

In the following set of equations ((3.4)- (3.9)) x represents a specfic aerodynamic surface of the aircraft. The total coeffcients (3.5) show how this blending occours.

$$\begin{aligned} C_{l_x}(\alpha_x) &= (1-\sigma_x)C_{l_{x_{airfroil}}}(\alpha_x) + \sigma_x C_{l_{flatplate}}(\alpha_x) \\ C_{d_x}(\alpha_x) &= (1-\sigma_x)C_{d_{x_{airfroil}}}(\alpha_x) + \sigma_x C_{d_{flatplate}}(\alpha_x) \end{aligned} \tag{3.5}$$

$$C_{l_{x_{airfroil}}}(\alpha_x) = C_{l0_x} + C_{l\alpha_x}(\alpha_x), \quad C_{d_{x_{airfroil}}}(\alpha_x) = C_{d0} + C_{d\alpha_x}(\alpha_x^2), \tag{3.6}$$

$$C_{l_{flatplate}}(\alpha) = 2sin(\alpha)cos(\alpha), \quad C_{d_{flatplate}}(\alpha) = 2sin^2(\alpha) \tag{3.7}$$

This model for the elevator accounts for the variation in force due to the elevator deflection by applying a modifier to the elevator angle of attack proportional to the deflection. The value of this proportional gain (τ) was fit log data from real world griffon pro flights.

$$\alpha_w = tan^{-1}\frac{\dot{z}}{\dot{x}} \qquad \alpha_e = \alpha_w + \epsilon\tau \tag{3.8}$$

$$\sigma_x(\alpha) = \frac{1+e^{-M_x(\alpha-t_{l_x})}+e^{M_x(\alpha-t_{u_x})}}{(1+e^{-M_x(\alpha-t_{l_x})})(1+e^{M_x(\alpha-t_{u_x})})} \tag{3.9}$$

The Resulting Coeffcients of the wing can be seen in Figure 3.2 where we have a continous function throughout $[-\pi, \pi]$

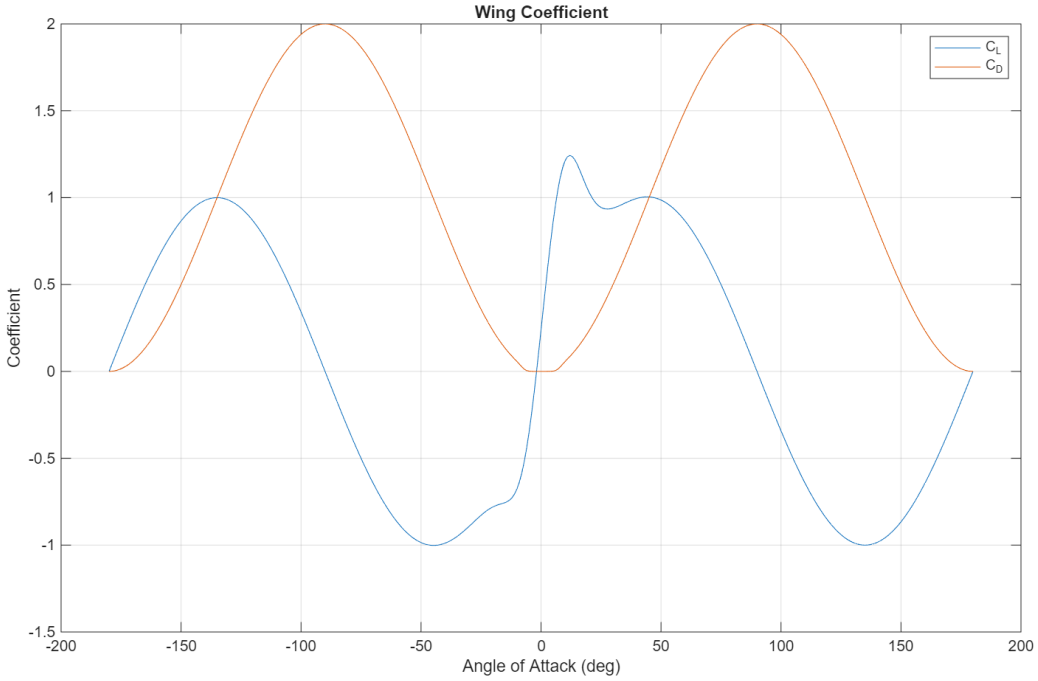


Figure 3.2: Griffon Pro Wing Aerodynamic Coeffcients

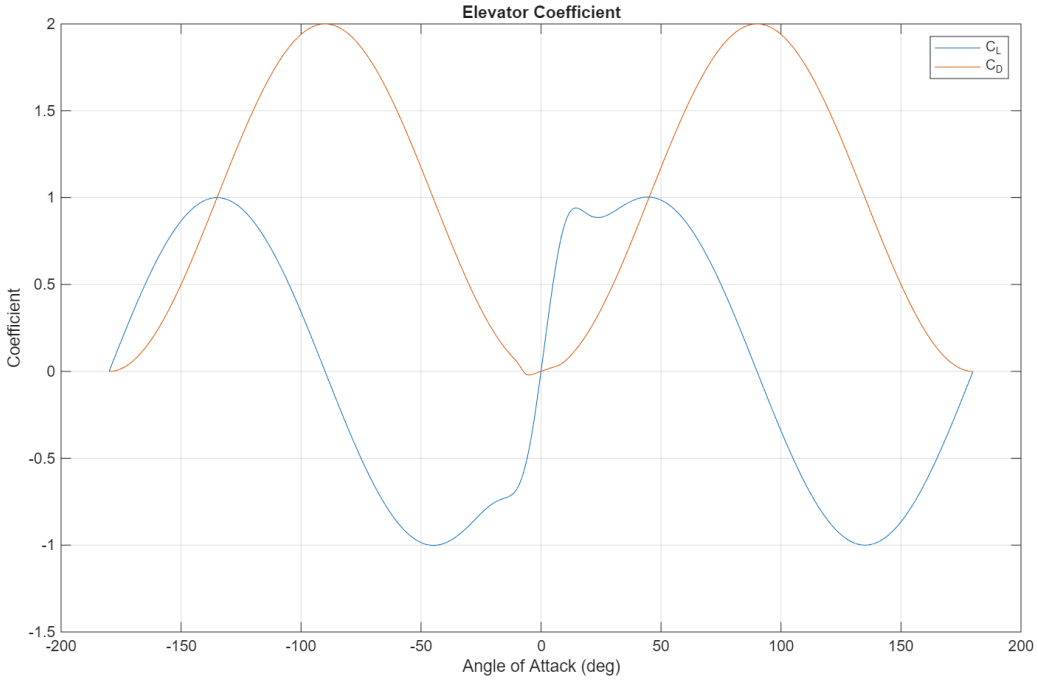


Figure 3.3: Griffon Pro Elevator Aerodynamic Coeffcients at $\epsilon = 0$

Coeffcient	Value	Coeffcient	Value
C_{l0_w}	1.9164	$C_{l\alpha_w}$	11.6632
C_{l0_e}	0	$C_{l\alpha_e}$	6.8755
C_{d0_w}	0.075	$C_{d\alpha_w}$	4.015
C_{d0_e}	0 .01	$C_{d\alpha_e}$	1.7968
t_{l_w}	-0.18	t_{u_w}	0.28
t_{l_e}	-0.28	t_{u_e}	0.28
M_w	1	M_e	26
τ	2.7		

Table 3.2: Aerodynamics Coefficients

3.1.2 Rotor Forces

The trajectory in section 3.2.6 utilises the non axial drag component the rotors produce when the freestream velocity has a perpendicular compenent to the rotor axle to dissipate kinetic energy from the aircraft, to model this behaviour the following equations were employed.

$$L_{front} = \frac{1}{2} N_b \rho c R_r^3 \omega_{front}^2 C_T, \qquad D_{front} = \frac{L_{front}(\dot{x}_{front})}{R_r \omega_{front}} \tag{3.10}$$

$$L_{rear} = \frac{1}{2} N_b \rho c R_r^3 \omega_{rear}^2 C_T, \qquad D_{rear} = \frac{L_{rear}(\dot{x}_{rear})}{R_r \omega_{rear}} \tag{3.11}$$

where N_b is the number of blades of the rotor, ρ is air density,c is the average chord of the airfoil, R_r is the radius of the rotor and C_T is the coefficient of thrust, it was estimated for hover by solving equation (3.13) which is the lift of the front and rear rotors equated to the weight force of the aircraft with the values from a real world flight log substituted for m, ρ, ω

$$\dot{X}_{front} = R_{IR}(\theta + \xi) X_{inertial}, \qquad \dot{X}_{rear} = R_{IR}(\theta + \xi) X_{inertial}, \qquad X = \begin{bmatrix} x \\ z \end{bmatrix} \tag{3.12}$$

$$C_T = \frac{2mg}{N_b \rho c R_r^3 (\omega_{front} + \omega_{rear})} \tag{3.13}$$

3.2 Trajectory Optimisation

The MATLAB library OptimTraj [17] was used to solve the optimisation problem for the transition trajectory. OptimTraj utilises MATLAB's fmincon solver.

3.2.1 Constraints

The constraints for the trajectory optimisation were as follows.

$$Y_{initial} = \begin{bmatrix} -400 & -50 & -0.0623 & 22 & 0 & 0 \end{bmatrix}^T, \qquad Y_{final} = \begin{bmatrix} 0 & -10 & 0 & 0 & 0 & 0 \end{bmatrix}^T \tag{3.14}$$

$$Y_{lower} = \begin{bmatrix} -\infty & -\infty & -\pi/10 & -\infty & \infty & -\pi/12 \end{bmatrix}^T, \qquad Y_{upper} = \begin{bmatrix} \infty & -10 & \pi/3 & 22 & \infty & \pi/12 \end{bmatrix}^T \tag{3.15}$$

$$U_{lower} = \begin{bmatrix} 0 & 0 & 0 & -\pi/4.75 \end{bmatrix}^T, \qquad U_{upper} = \begin{bmatrix} 8500 & 8500 & \pi/2 & \pi/4.75 \end{bmatrix}^T \tag{3.16}$$

$$11\pi/90 - \alpha >= 0, \qquad \alpha - \pi/18 >= 0 \tag{3.17}$$

The control constraints are determined by the capabilities of the aircraft, with the exception of the tilt angle (ξ) a 15% reduction to the upper limit was applied to ensure there was adequate control margin. Y_{final} is set to a stationary hover 10m above the ground as to allow the stock ArduPilot to take over for the touch down and complete its disarming protocols uninterrupted, this height can also be altered along the state upper constraint for altitude to set a minimum height that the pilot feels is appropriate. The final constraint was applied to the angle of attack of the airframe, to limit aggressiveness of the stalls the aircraft could perform, this too can be configured to a value the pilot deems appropriate.

3.2.2 Initialisation

The initialisation point of the optimisation was a straight line trajectory with the inputs at zero for all time.

$$Y_{initialisation} = \begin{bmatrix} Y_{initial}, & Y_{final} \end{bmatrix}, U_{initialisation} = 0^{\mathbb{R}^{4\times 2}} \tag{3.18}$$

3.2.3 Mesh Refinement

The optimisation mesh would initially be calculated with 0.5s timesteps (Δt), and time would be a free variable. After the initial solution had converged, the total time would be rounded to the nearest 0.5s and be set as a constraint and the timesteps would be reduced to 0.25s, with the initial solution replacing the previous initialisations for state and control input. This process was repeated until a time step of 0.1s was achieved.

3.2.4 Cost Function

The goal of this optimisation is to find the most energy efficient landing transition trajectory, as such the cost function was set to the total energy consumed during the transition phase, with the assumption that elevator and tilt inputs would cause marginal differences the cost function was therefore simplified to be the energy the rotors require throughout the trajectory.

$$J_{trajectory} = \sum_{i=0}^{\frac{t_f}{\Delta t}} 2(P_{front_i} + P_{rear_i})\Delta t \tag{3.19}$$

The power estimates show in 3.19 are derived using the method shown in [22], and can also be found in Appendix ??.

3.2.5 Solution

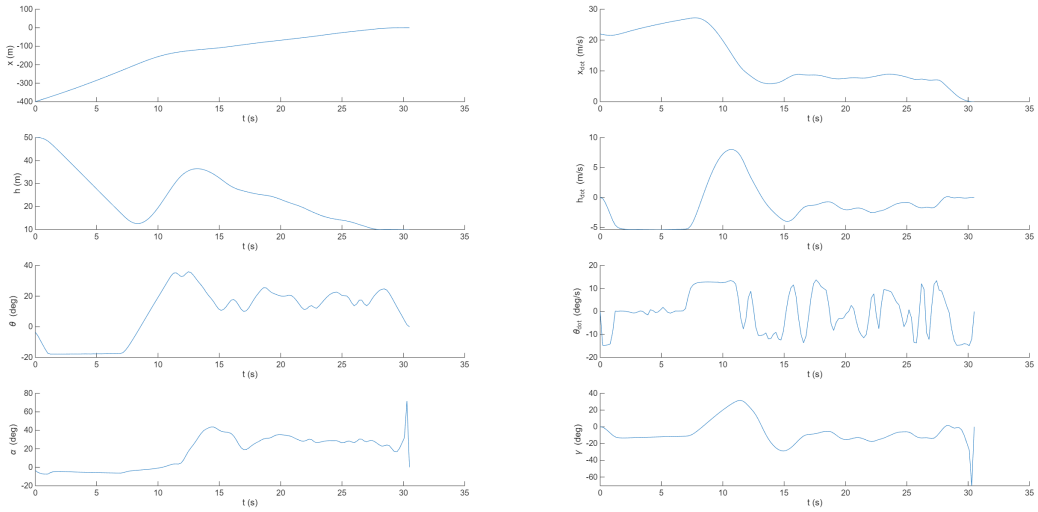


Figure 3.4: Trajectory Optimisation Solution States

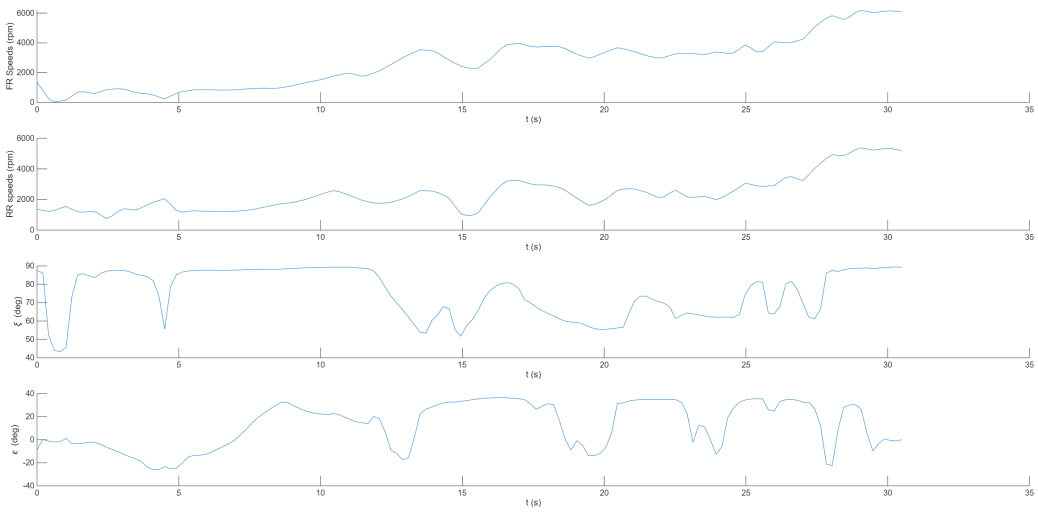


Figure 3.5: Trajectory Optimisation Solution Control Inputs

3.3 Control

The control of the aircraft is achieved through a time varying linear quadratic regulator (LQR), utilising the optimisation solution as the feedforward term for the longitudinal control and a PID architecture for the lateral stabilisation.

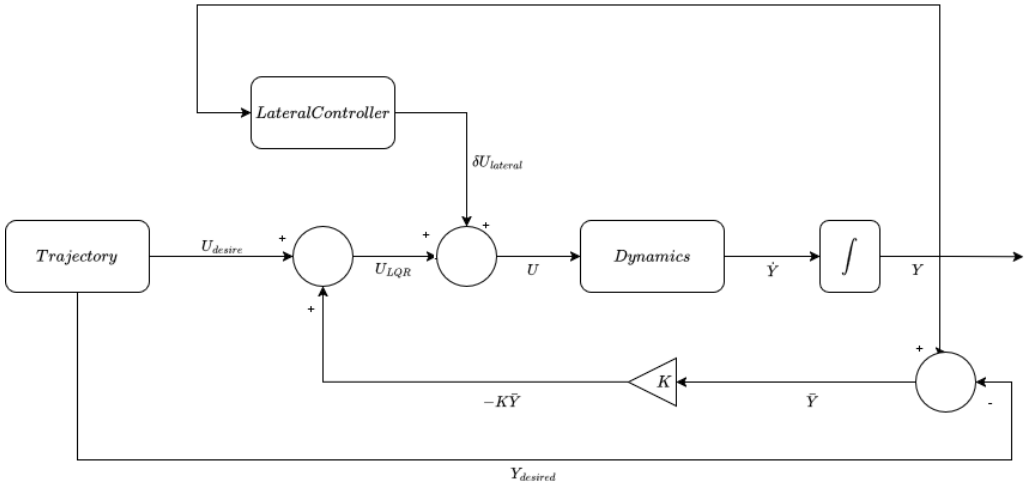


Figure 3.6: Longitudinal Control Block Diagram

The choice of LQR for the longitudinal dynamics was driven by the desire for an efficient control architecture with respect to the goal of this project minimising control energy and to minimise the computation required in real time on the flight controller, as the gains can be precomputed.

3.3.1 LQR

The LQR controller only controls the longitudinal dynamics as shown in section 3.2, therefore $Y_d \in \mathbb{R}^{6\times\frac{t_f}{\Delta t}}$ and $U_d \in \mathbb{R}^{4\times\frac{t_f}{\Delta t}}$ this is then linearly interpolated to produce $y_d(t) \in \mathbb{R}^6$ and $u_d(t) \in \mathbb{R}^4$. with these continuous vectors equation (3.20) can be formed, where $\bar{y}(t) = y(t) - y_d(t)$ and $\bar{u}(t) = u(t) - u_d(t)$

$$\dot{\bar{y}}(t) = f(\bar{y}(t), \bar{u}(t), t) \tag{3.20}$$

As the dynamics of the system are non-linear it was linearised using the Jacobians: $A(t) = \frac{\partial f(t)}{\partial y(t)}$, $B(t) = \frac{\partial f(t)}{\partial u(t)}$ to form equation (3.21), with a timestep of $1 \times 10^{-4}s$

$$\dot{\bar{y}}(t) = A(t)\bar{y}(t) + B(t)\bar{u}(t) \tag{3.21}$$

3.3.1.1 LQR Gains

The LQR Controller calculates the gains by solving the following cost function:

$$J_{LQR} = \int_0^{\frac{t_f}{\Delta t}} y^T(t)Qy(t) + u^T(t)Ru(t)\,dt \tag{3.22}$$

where $Q \in \mathbb{R}^{6\times6}$ is the state error cost matrix and $R \in \mathbb{R}^{4\times4}$ is the control cost matrix. Both were set using Bryson's rule, stated below.

$$Q_{ii} = \frac{1}{\bar{y}^2_{i_{max}}}, R_{ii} = \frac{1}{\bar{u}^2_{i_{max}}} \tag{3.23}$$

resulting in the following values for the Griffin Pro: $Q = diag(4, 4, 14.5903, 1.111 \times 10^{-}3, 25, 8.2070)$ and $R = diag(4 \times 10^{-6}, 4 \times 10^{-6}, 1.3131 \times 10^{4}, 1.4590 \times 10^{3})$. With Q and R selected the differential Riccati equations can be solved moving back from the terminal point of the trajectory assuming zero final state and input error, to find the gain matrices.

$$-\dot{S}(t) = A^T(t)S(t) + S(t)A(t) - S(t)B(t)R^{-1}B^T(t)S(t) + Q \tag{3.24}$$

$$K(t) = R^{-1}B^T(t)S(t) \tag{3.25}$$

At this point $K \in \mathbb{R}^{4\times 6t_f 10000}$ is this much larger than the available memory on most flight controllers and therefore every $10,000^{th}$ term is kept and the rest were discarded forming a new matrix. This is the extent of the precomputation achieved, next $Y_d \in \mathbb{R}^{6\times t_f \Delta t}$, $U_d \in \mathbb{R}^{4\times t_f \Delta t}$, and $K_{reduced} \in \mathbb{R}^{4\times 6t_f}$ were imported into the ArduPilot Lua scripting environment.

Within the Lua script the control inputs are converted from their native units (Table 3.1) to μs which represents the pulse duration in the aircraft esc. This utilises the ArduPilot function *"SRV_Channels:setoutput_pwm_chan_timeout"*. The elevator and tilt servos are converted by a linear map, however the motor rpm to μs a linear map is employed followed by a SISO PD controller on

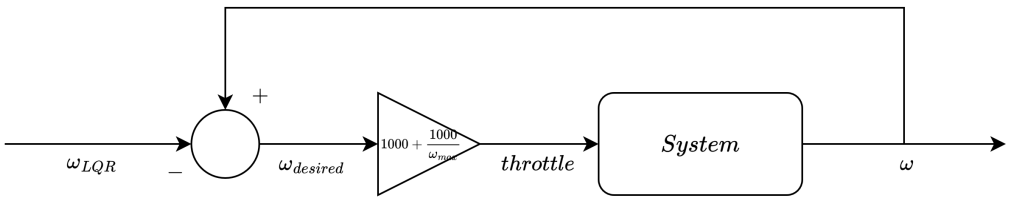


Figure 3.7: Rotor Control Block Diagram

3.3.2 Lateral Stabilisation

As the LQR models is limited to the longitudinal dynamics the lateral stabilisation is a non-optimal series of PD controllers, encompassing roll, yaw, y axis position, controlled through modifiers to the LQR control inputs for the motors and tilt servos, as shown in figure 3.4.

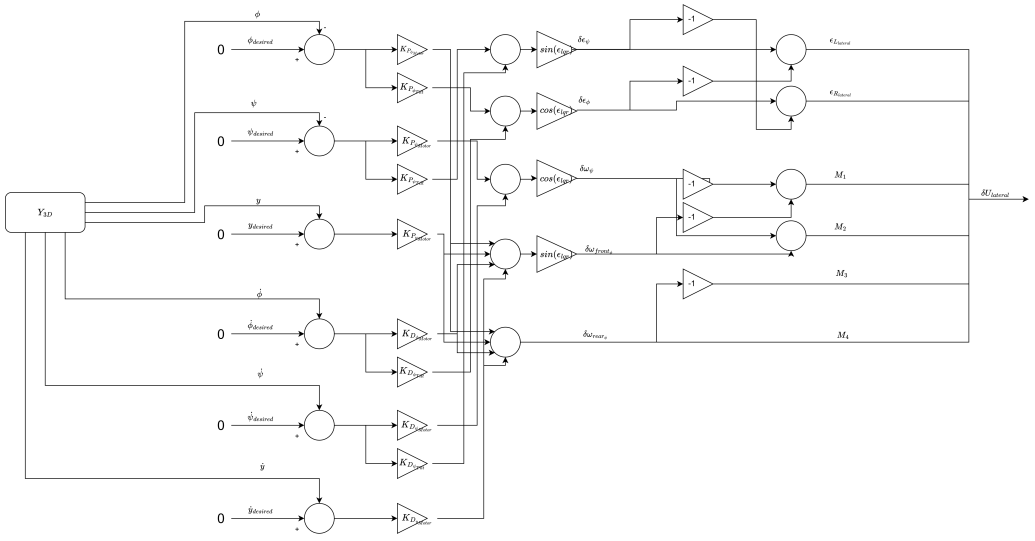


Figure 3.8: Lateral Control Block Diagram

The gains for these PD controllers were tuned with the aid of a large language model agent running simulation in the loop (SITL) runs until the state errors were below the required threshold: $\bar{Y}_{3D} = [0.1745, 0.1745, 1, 0.0524, 0.0524, 0.5]^T|$ resulting in the gains below.

$$a = 0.2\Delta t_{lua} + 1 \tag{3.26}$$

Gain	Value	Unit	Gain	Value	Unit
$K_{P_{\phi_{Motor}}}$	$\frac{300}{a}$	$rad- > \mu s$	$K_{D_{\phi_{Motor}}}$	$\frac{50}{a}$	$rad/s- > \mu s$
$K_{P_{\psi_{Motor}}}$	$\frac{600}{a}$	$rad- > \mu s$	$K_{D_{\psi_{Motor}}}$	$\frac{100}{a}$	$rad/s- > \mu s$
$K_{P_{y_{Motor}}}$	$\frac{60}{a}$	$m- > \mu s$	$K_{D_{\phi_{Motor}}}$	$\frac{120}{a}$	$m/s- > \mu s$
$K_{P_{\phi_\xi}}$	$\frac{0.2}{a}$	$rad- > \mu s$	$K_{D_{\phi_\xi}}$	$\frac{0.03}{a}$	$rad/s- > \mu s$
$K_{P_{\phi_\xi}}$	$\frac{0.3}{a}$	$rad- > \mu s$	$K_{D_{\phi_\xi}}$	$\frac{0.05}{a}$	$rad/s- > \mu s$

Table 3.3: Lateral Controller Gains

where Δt_{lua} is the time step between each cycle of the lua script. Each of the five PD controllers is individually clamped, the max tilt alteration is 15 deg and $200\mu s$ for the pwm signals. The layering of the LQR and lateral controllers means that the stability of the system is not guaranteed, as such the poles of the full system were calculated. Figure 3.6 shows the maximum real value of the poles throughout the landing approach.

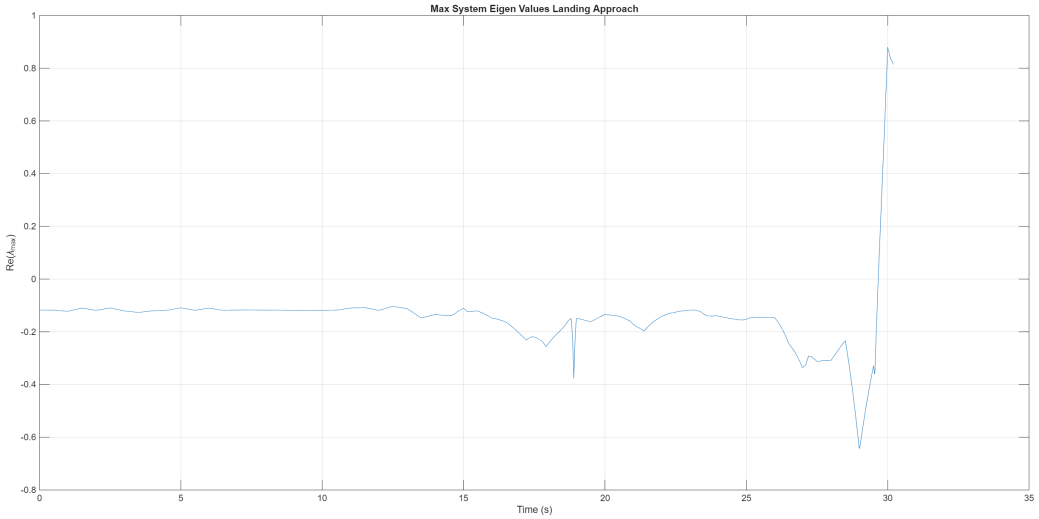


Figure 3.9: LQR Re(λ_{max})

The max real pole is negative throughout the approach ensuring stability until $t \approx 29.66s$. This is due to the LQR cost function minimising both state error and input cost, meaning that the LQR gains wind-down as the system approaches the time horizon because the error doesn't grow enough before ending to warrant the cost incurred by the rotors.

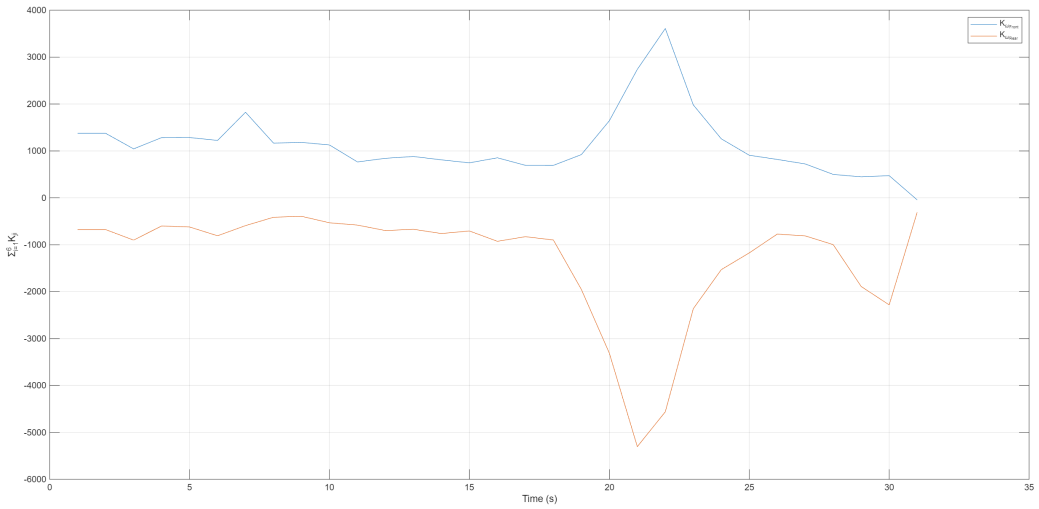


Figure 3.10: LQR Rotor Gains

3.4 Simulation Enviroment

3.4.1 Rotor Model

The public release of ArduPilot's simulation environment doesn't model the non-axial forces of the rotors and therefore I have modified the simulator to model the rotors utilising momentum theory. These changes to the rotor model work with any controller or aircraft running ArduPilot and thus other may wish to use it in their own simulations. The implementations of these changes follow the same approach as those shown in section 3.1.2, with a varition in how the velocity is the drag term is calculated to work in the three dimenstional simulation.

$$\dot{X}_{front} = R_{IR}(\phi, \theta + \xi - \frac{\pi}{2}, \psi), \qquad \dot{X}_{rear} = R_{IR}(\phi, \theta + \xi, \psi), \qquad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \tag{3.27}$$

3.5 Safety Features

3.5.1 Altitude Floor

The Altitude floor is a minimum altitude that if exceeded will cease the control input overrides, returning control to the stock ArduPilot controller. Controller wind up is a concern for the stock controller at handoff, however, from simulation testing it has not been a problem, reference 3.11.

3.5.2 State Error Disconnect

Another method for rejecting the landing sequence is if the norm of the state error surpasses a preset limit. This triggers in two cases, the first is the aircraft failing to track the trajectory (refer to 3.11), the second being a mismatch in the initial condition for example if the trajectory had not been updated and was made for a different altitude approach.

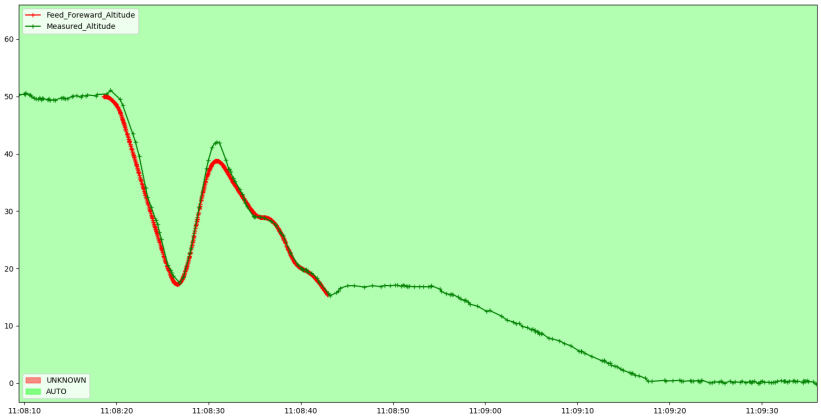


Figure 3.11: Altitude Floor Rejected Landing (15m)

Results and Analysis

With the application of model based optimal control the energy consumption of a Arace Griffin Pro UAV during landing approach has been reduced by 61%, This can improve efficiency, payload and range, without the need for any new hardware. This approach could be broadened to apply to all powered lift aircraft running ArduPilot.

Figure 4.1 shows that the primary reduction in energy consumption comes from the lack time spent in the hover phase (xs to ys) as the Lua controller only hovers down 10m instead of the stock 50m, with the hover portion of the approach able to be reduced even further by changing the optimisation constraints in section 3.2.2 to achieve even higher efficiency, if the operator is comfortable with the increased risk.

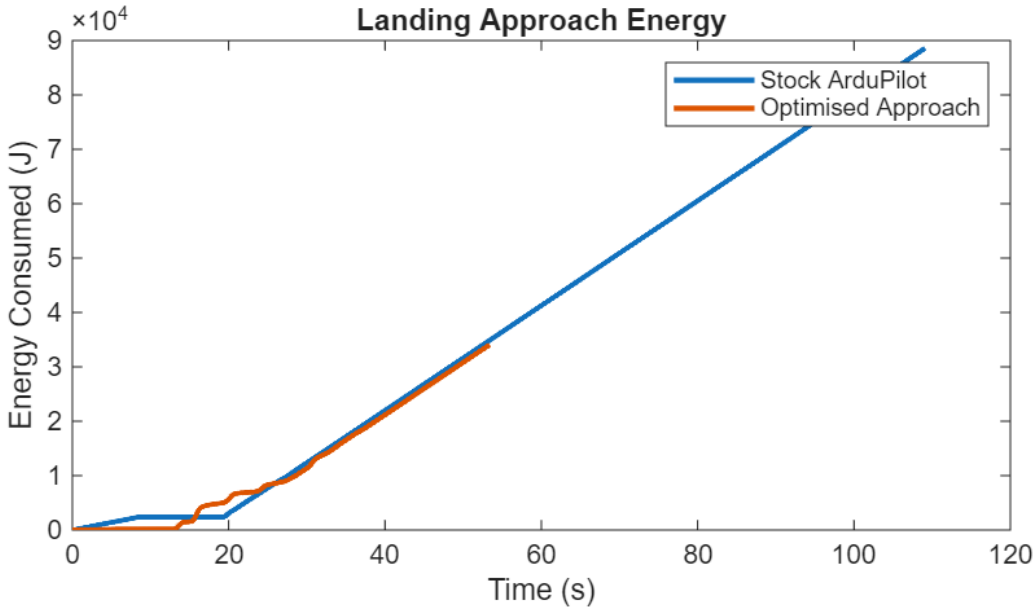


Figure 4.1: Landing Approach Energy Consumption

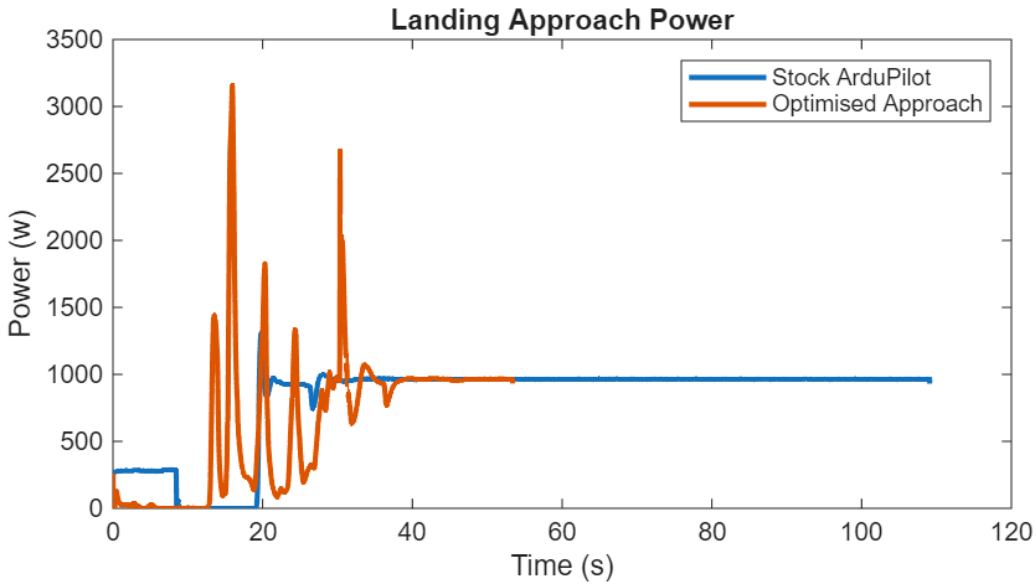
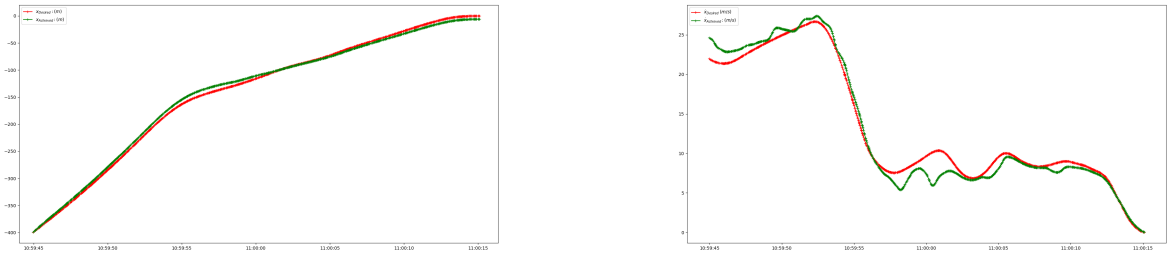


Figure 4.2: Landing Approach Power

The controller prioritises tracking the positions over the rates as seen in figures 4.3 through 4.10. on the basis of a small error in climb rate is less important than a error in altitude when diving towards the ground as seen in first 8s of the landing approach.

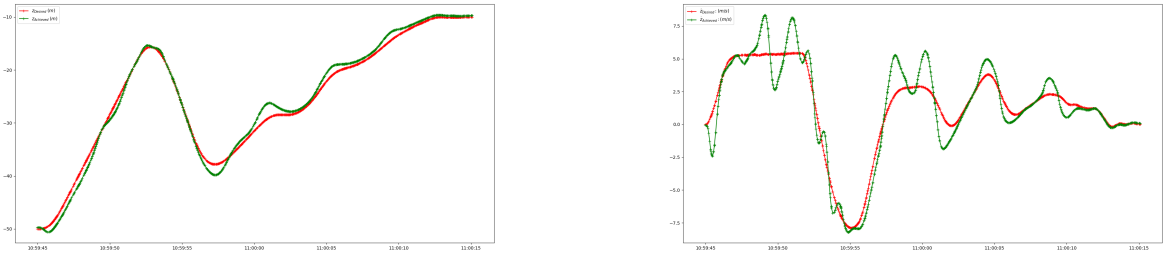
Further improvement could be made to the lateral control by incorporating the rudder and aileron into the script controller, this would also have the added benefit of relaxing the constraints on the waypoints in ArduPilot, as the current solution requires a waypoint that lies on the line drawn between the VTOL Landing and the previous waypoint to be the trajectory $x[0] - waypoint_{radius}$ from the landing waypoint. if it lies off the line the ArduPilot driven control surfaces, primarily the ailerons will drive the aircraft off track.

The other benefit of incorporating the full lateral dynamics would be that the lateral control could be achieved through the LQR which would provide a more optimal solution to the disturbance rejection control. Additionally if the lateral dynamics were integrated this controller would capable of driving any arbitrary trajectory, widening the possible applications.



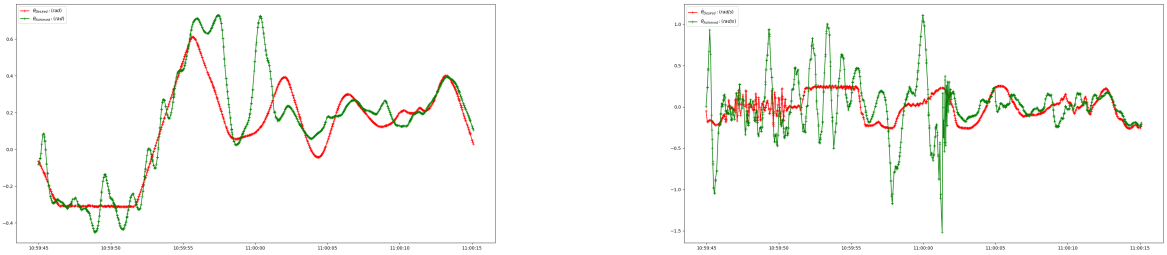
(a) x (b) $\dot{x}$

Figure 4.3: Simulation x and $\dot{x}$



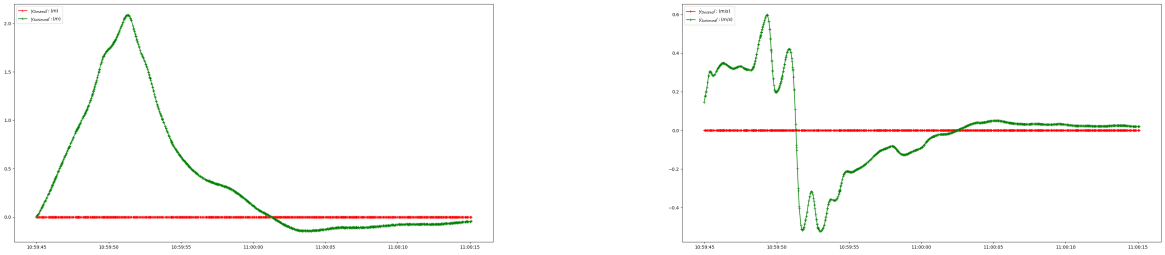
(a) z (b) $\dot{z}$

Figure 4.4: Simulation z and $\dot{z}$



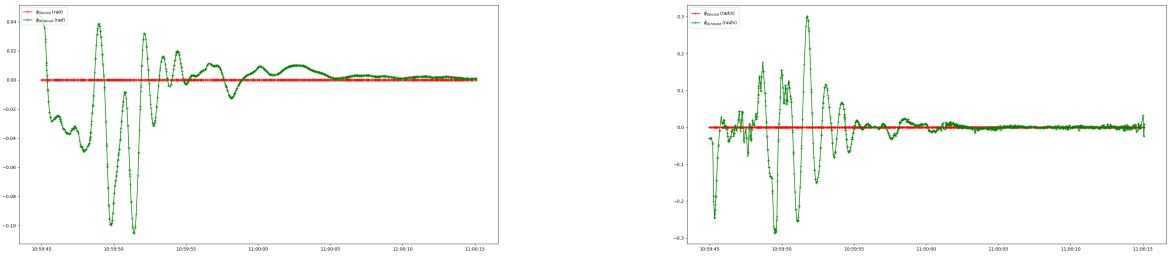
(a) θ (b) $\dot{\theta}$

Figure 4.5: Simulation θ and $\dot{\theta}$



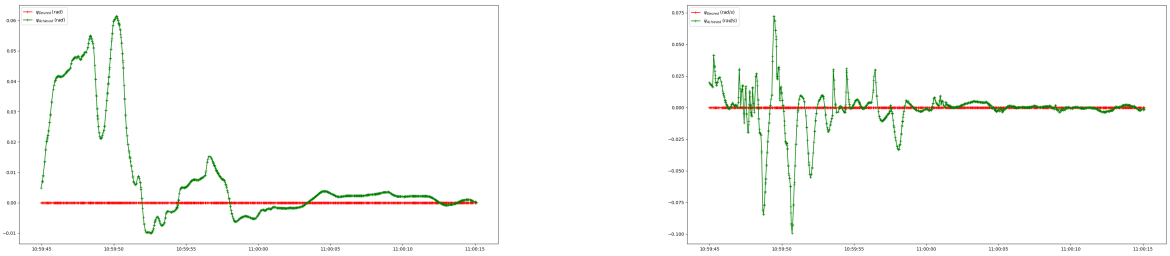
(a) y (b) $\dot{y}$

Figure 4.6: Simulation y and $\dot{y}$



(a) ϕ (b) $\dot{\phi}$

Figure 4.7: Simulation ϕ and $\dot{\phi}$



(a) ψ (b) $\dot{\psi}$

Figure 4.8: Simulation ψ and $\dot{\psi}$

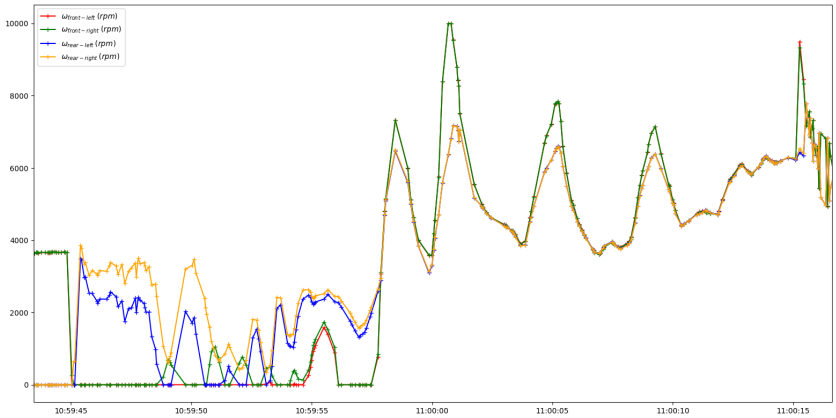
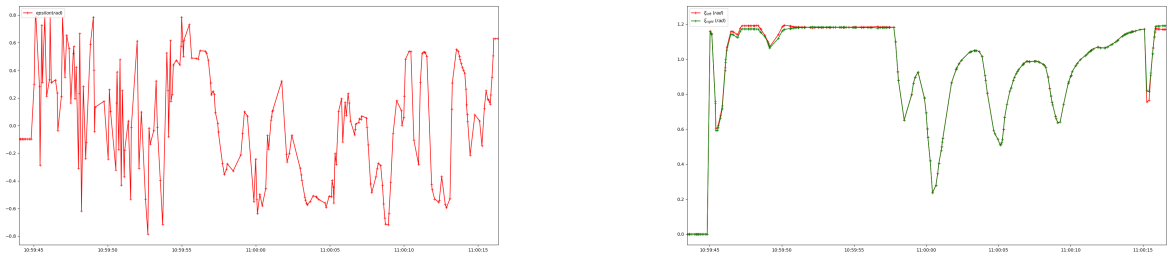


Figure 4.9: Simulation ω



(a) ϵ (b) ξ

Figure 4.10: Simulation ϵ and ξ

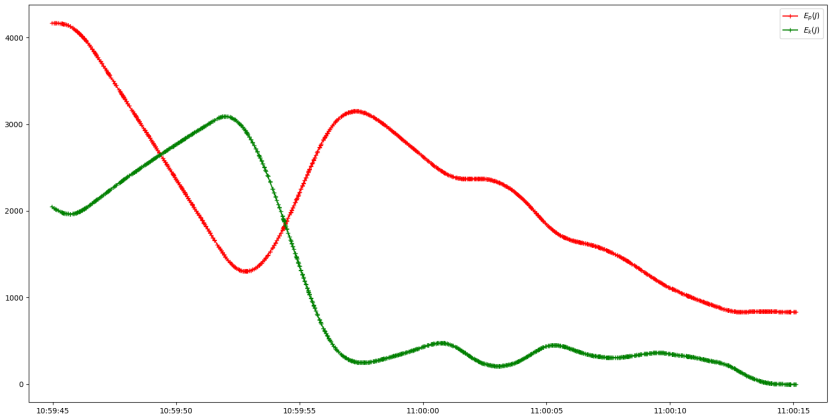


Figure 4.11: Landing Approach Energy States

The approach trajectory has two objectives of bleeding off the kinetic and potential energy of the aircraft. The solution immediately converts the potential energy to kinetic through the dive in the first 8s of the approach, followed by a pull up into a stall, that hits the pitch rate constraint causing the gain in height, however this stall also dissipates the vast majority of the kinetic energy, which remains largely constant in final stage of the approach (12s-30.5s) as the aircraft holds the upper angle of attack constraint (20 deg) and therefore a stall stabilised with the quad rotors to slowly approach the hover for handover.

Conclusions and Future Development

5.1 Conclusion

The project so far has been a success in reducing the energy consumption of the aircraft during the landing transition achieving a 61% reduction, with the added benefit of a substantial time reduction, however as the results are only simulations real world testing is required to fully validate these results.

5.2 Future Development

The work put forward in this project while promising is not complete, and there are several areas of future work on this topic worth pursuing.

5.2.1 Validation

This project has only presented results from simulation. Real world flight testing is required to validate the control with the extra latency associated with the microcontroller, any model mismatch between the dynamical model presented in section 3.1 and real environmental disturbances such as wind, which have been simulated (refer to appendix XXX for hover gust response simulations.).

5.2.2 Model Expansion

While the initial decision was taken to limit the dynamical model to the longitudinal frame to reduce complexity, it now seems prudent to incorporate the lateral dynamics, as it would allow the lateral stabilisation to be incorporated into the LQR controller rather than the current PD solution, as well as allowing for three dimensional approaches, which would mitigate one of the limitations of the current trajectories requiring a long clear lane to approach the landing point.

5.2.3 Airframe types

This project in its current state exclusively supports tilt rotor powered lift UAVs, however, the same approach could be applied to tail sitters, tilt wings and quadplanes without excessive effort, allowing all powered lift aircraft to reap the benefits of this work.

5.2.4 System Identification

A dynamical model is required for both the trajectory optimisation and LQR control as shown in section 3, however to build this model a large number of parameters of the aircraft are required with the most difficult to measure being the moments of inertia around the center of gravity. As such a

system identification protocol would be a valuable addition to this work to provide users an estimate of these values especially if the model was extended to three dimensions.

Appendix A: The First

Manage your references and present them here in order.

Appendix B: The Second

Manage your references and present them here in order.

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